Readme

1. This Appendix presents the validation process with Laplace's equation for second- and third-order potential gradients in Tables A1-A4 and 1-2.
2. Each of the following sections is independent by using the command **Clear["Global`*"]** to clear previous arguments.
3. After running all codes, the results of Laplace's equation are obtained as zero.

1. Table A1

```
In[•]:= Clear["Global`*"];
```

$$\text{In}[\bullet]:= l = \sqrt{r^2 + r3^2 - 2\, r\, r3\, \text{Cos}[\theta 3]}\;;\; A = l + r3 - r\, \text{Cos}[\theta 3];\; U = \frac{\sqrt{r^2 + r3^2 - 2\, r\, r3\, \text{Cos}[\theta 3]}}{\sqrt{r^2 + r3^2}};$$

$$\text{In}[\bullet]:= VxxP = \frac{G\rho}{48}\left(-3\,\text{Cos}[\theta 3]\,\left(8\,(\lambda 2-\lambda 1)\,\text{Sin}[\theta 3]^2 + 4\,\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1]\,(5-\text{Cos}[2\theta 3])\right)\right.$$
$$\text{Log}[A] + \frac{1}{r^3\, l}\left(-4\left(r^4 + r^2\, r3^2 + 4\, r3^4\right)(\lambda 2-\lambda 1) + 2\, r^2\left(\left(3\, r^2 + r3^2\right)\text{Cos}[2\theta 3] - 3\, r\, r3\,\text{Cos}[3\theta 3]\right)(2\lambda 2 - 2\lambda 1 + \text{Sin}[2(\lambda-\lambda 1)] - \text{Sin}[2(\lambda-\lambda 2)]) - 4\left(13\, r^4 + 13\, r^2\, r3^2 + 4\, r3^4\right)\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1] + 4\, r\, r3\,\text{Cos}[\theta 3]\left(\left(r^2 + 4\, r3^2\right)(\lambda 2-\lambda 1) + \left(25\, r^2 + 4\, r3^2\right)\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1]\right)\right)\Bigg);$$

$$\text{In}[\bullet]:= VyyP = \frac{G\rho}{48}\left(-3\,\text{Cos}[\theta 3]\,\left(8\,(\lambda 2-\lambda 1)\,\text{Sin}[\theta 3]^2 - 4\,\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1]\,(5-\text{Cos}[2\theta 3])\right)\right.$$
$$\text{Log}[A] + \frac{1}{r^3\, l}\left(-4\left(r^4 + r^2\, r3^2 + 4\, r3^4\right)(\lambda 2-\lambda 1) + 2\, r^2\left(\left(3\, r^2 + r3^2\right)\text{Cos}[2\theta 3] - 3\, r\, r3\,\text{Cos}[3\theta 3]\right)(2\lambda 2 - 2\lambda 1 - \text{Sin}[2(\lambda-\lambda 1)] + \text{Sin}[2(\lambda-\lambda 2)]) + 4\left(13\, r^4 + 13\, r^2\, r3^2 + 4\, r3^4\right)\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1] + 4\, r\, r3\,\text{Cos}[\theta 3]\left(\left(r^2 + 4\, r3^2\right)(\lambda 2-\lambda 1) - \left(25\, r^2 + 4\, r3^2\right)\text{Cos}[2\lambda-\lambda 1-\lambda 2]*\text{Sin}[\lambda 2-\lambda 1]\right)\right)\Bigg);$$

$$\text{In}[\bullet]:= VzzP = \frac{G\rho\,(\lambda 2-\lambda 1)}{6\, r^3}\left(\frac{1}{l}\left(r^4 + r^2\, r3^2 + 4\, r3^4 - r\, r3\left(r^2 + 4\, r3^2\right)\text{Cos}[\theta 3] - r^2\left(3\, r^2 + r3^2\right)\text{Cos}[2\theta 3] + 3\, r^3\, r3\,\text{Cos}[3\theta 3]\right) + 6\, r^3\,\text{Sin}[\theta 3]^2\,\text{Cos}[\theta 3]*\text{Log}[A]\right);$$

```
In[•]:= VxxzP = G * ρ / (16 * r^4 * l^3)
    (6 * r^2 * r3^4 * Cos[2 * θ3] * (2 * λ2 - 2 * λ1 + Sin[2 * (λ - λ1)] - Sin[2 * (λ - λ2)]) -
      Cos[2 * λ - λ1 - λ2] * Sin[λ2 - λ1] *
       (5 * r^6 * U + r3^6 * (5 * U - 16) + r^2 * r3^4 * (15 * U - 52) + r^4 * r3^2 * (15 * U - 16)) + 2 * r *
      r3 * Cos[θ3] * ((λ2 - λ1) * (5 * r^4 * U + r3^4 * (5 * U - 24) + 2 * r^2 * r3^2 * (5 * U - 4)) +
         (r^2 + r3^2) * Cos[2 * λ - λ1 - λ2] * Sin[λ2 - λ1] * (5 * r^2 * U + r3^2 * (5 * U - 24))) -
      5 (r^2 + r3^2)^3 * (λ2 - λ1) * U + 4 * r3^4 * (9 * r^2 + 4 * r3^2) * (λ2 - λ1));

In[•]:= VyyzP = G * ρ / (16 * r^4 * l^3)
    (6 * r^2 * r3^4 * Cos[2 * θ3] * (2 * λ2 - 2 * λ1 - Sin[2 * (λ - λ1)] + Sin[2 * (λ - λ2)]) +
      Cos[2 * λ - λ1 - λ2] * Sin[λ2 - λ1] *
       (5 * r^6 * U + r3^6 * (5 * U - 16) + r^2 * r3^4 * (15 * U - 52) + r^4 * r3^2 * (15 * U - 16)) + 2 * r *
      r3 * Cos[θ3] * ((λ2 - λ1) * (5 * r^4 * U + r3^4 * (5 * U - 24) + 2 * r^2 * r3^2 * (5 * U - 4)) -
         (r^2 + r3^2) * Cos[2 * λ - λ1 - λ2] * Sin[λ2 - λ1] * (5 * r^2 * U + r3^2 * (5 * U - 24))) -
      5 (r^2 + r3^2)^3 * (λ2 - λ1) * U + 4 * r3^4 * (9 * r^2 + 4 * r3^2) * (λ2 - λ1));

In[•]:= VzzzP = -(G * ρ * (λ2 - λ1)) / (2 * r^4 * l^3)
    (r3^3 * (9 * r^2 * r3 + 4 * r3^3 - 4 * r * (r^2 + 3 * r3^2) * Cos[θ3] + 3 * r^2 * r3 * Cos[2 * θ3]));

In[•]:= FullSimplify[VxxP + VyyP + VzzP]
Out[•]=
    0

In[•]:= FullSimplify[VxxzP + VyyzP + VzzzP]
Out[•]=
    5 G (r^2 + r3^2)^(3/2) (λ1 - λ2) ρ / (8 r^4)

In[•]:= f[θ3_] := 5 G (r^2 + r3^2)^(3/2) (λ1 - λ2) ρ / (8 r^4); f[θ2] - f[θ1]
Out[•]=
    0
```

2. Table A2

```
In[•]:= Clear["Global`*"];

In[•]:= l = √(r^2 + r3^2 - 2 r r3 Cos[θ3]); A = l + r3 - r Cos[θ3]; U = √(r^2 + r3^2 - 2 r r3 Cos[θ3]) / √(r^2 + r3^2);

In[•]:= VxxZB = -(Pi G ρ / (6 r^3)) ((1/l) (r^4 + r^2 r3^2 + 4 r3^4 - r r3 (r^2 + 4 r3^2) Cos[θ3] -
        r^2 (3 r^2 + r3^2) Cos[2 θ3] + 3 r^3 r3 Cos[3 θ3]) + 6 r^3 Sin[θ3]^2 * Cos[θ3] * Log[A]);
```

```
In[•]:= VyyZB = -(Pi G ρ)/(6 r^3) ((1/l)(r^4 + r^2 r3^2 + 4 r3^4 - r r3 (r^2 + 4 r3^2) Cos[θ3] -
          r^2 (3 r^2 + r3^2) Cos[2 θ3] + 3 r^3 r3 Cos[3 θ3]) + 6 r^3 Sin[θ3]^2 * Cos[θ3] * Log[A]);

In[•]:= VzzZB = (Pi G ρ)/(3 r^3) ((1/l)(r^4 + r^2 r3^2 + 4 r3^4 - r r3 (r^2 + 4 r3^2) Cos[θ3] -
          r^2 (3 r^2 + r3^2) Cos[2 θ3] + 3 r^3 r3 Cos[3 θ3]) + 6 r^3 Sin[θ3]^2 * Cos[θ3] * Log[A]);

In[•]:= VxxzZB = (Pi G ρ)/(8 r^4 l^3) (4 * r3^4 * (9 * r^2 + 4 * r3^2 + 3 * r^2 * Cos[2 θ3]) + 2 * r * r3 * Cos[θ3] *
          (5 * r^4 * U + r3^4 * (5 * U - 24) + 2 * r^2 * r3^2 * (5 * U - 4)) - 5 * (r^2 + r3^2)^3 * U);

In[•]:= VyyzZB = (Pi G ρ)/(8 r^4 l^3) (4 * r3^4 * (9 * r^2 + 4 * r3^2 + 3 * r^2 * Cos[2 θ3]) + 2 * r * r3 * Cos[θ3] *
          (5 * r^4 * U + r3^4 * (5 * U - 24) + 2 * r^2 * r3^2 * (5 * U - 4)) - 5 * (r^2 + r3^2)^3 * U);

In[•]:= VzzzZB = -(Pi G ρ r3^3 (9 r^2 r3 + 4 r3^3 - 4 r (r^2 + 3 r3^2) Cos[θ3] + 3 r^2 r3 Cos[2 θ3]))/(r^4 l^3);

In[•]:= FullSimplify[VxxZB + VyyZB + VzzZB]
Out[•]=
0

In[•]:= FullSimplify[VxxzZB + VyyzZB + VzzzZB]
Out[•]=
-(5 G π (r^2 + r3^2)^(3/2) ρ)/(4 r^4)

In[•]:= g[θ3_] := -(5 G π (r^2 + r3^2)^(3/2) ρ)/(4 r^4); g[θ2] - g[θ1]
Out[•]=
0
```

3. Table A3

```
In[•]:= Clear["Global`*"];

In[•]:= l1 = Sqrt[r^2 + r1^2 - 2 r r1 Cos[θ3]]; l2 = Sqrt[r^2 + r2^2 - 2 r r2 Cos[θ3]]; B = (l1 + l2 + (r2 - r1))/(l1 + l2 - (r2 - r1));
```

```
In[•]:= VxxP = G ρ/48
          (-3 Cos[θ3] (8 (λ2 - λ1) Sin[θ3]^2 + 4 Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1] (5 - Cos[2 θ3]))
            Log[B] + 1/(r^3 l) (-4 (r^4 + r^2 r3^2 + 4 r3^4) (λ2 - λ1) + 2 r^2 ((3 r^2 + r3^2) Cos[2 θ3] -
                  3 r r3 Cos[3 θ3]) (2 λ2 - 2 λ1 + Sin[2 (λ - λ1)] - Sin[2 (λ - λ2)]) -
              4 (13 r^4 + 13 r^2 r3^2 + 4 r3^4) Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1] + 4 r r3 Cos[θ3]
               ((r^2 + 4 r3^2) (λ2 - λ1) + (25 r^2 + 4 r3^2) Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1])));

In[•]:= VyyP = G ρ/48
          (-3 Cos[θ3] (8 (λ2 - λ1) Sin[θ3]^2 - 4 Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1] (5 - Cos[2 θ3]))
            Log[B] + 1/(r^3 l) (-4 (r^4 + r^2 r3^2 + 4 r3^4) (λ2 - λ1) + 2 r^2 ((3 r^2 + r3^2) Cos[2 θ3] -
                  3 r r3 Cos[3 θ3]) (2 λ2 - 2 λ1 - Sin[2 (λ - λ1)] + Sin[2 (λ - λ2)]) +
              4 (13 r^4 + 13 r^2 r3^2 + 4 r3^4) Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1] + 4 r r3 Cos[θ3]
               ((r^2 + 4 r3^2) (λ2 - λ1) - (25 r^2 + 4 r3^2) Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1])));

In[•]:= VzzP =
          G ρ (λ2 - λ1)/(6 r^3) (1/l (r^4 + r^2 r3^2 + 4 r3^4 - r r3 (r^2 + 4 r3^2) Cos[θ3] - r^2 (3 r^2 + r3^2) Cos[2 θ3] +
                3 r^3 r3 Cos[3 θ3]) + 6 r^3 Sin[θ3]^2 Cos[θ3] * Log[B]);

In[•]:= FullSimplify[VxxP + VyyP + VzzP]
Out[•]=
        0
```

4. Table A4

```
In[•]:= Clear["Global`*"];

In[•]:= l = √(r^2 + r3^2 - 2 r r3 Cos[θ3]);
        l1 = √(r^2 + r1^2 - 2 r r1 Cos[θ3]);
        l2 = √(r^2 + r2^2 - 2 r r2 Cos[θ3]);
        B = (l1 + l2 + (r2 - r1))/(l1 + l2 - (r2 - r1));

In[•]:= VxxZB = -Pi G ρ/(6 r^3) (1/l (r^4 + r^2 r3^2 + 4 r3^4 - r r3 (r^2 + 4 r3^2) Cos[θ3] -
              r^2 (3 r^2 + r3^2) Cos[2 θ3] + 3 r^3 r3 Cos[3 θ3]) + 6 r^3 Sin[θ3]^2 * Cos[θ3] * Log[B]);
```

```
In[•]:= VyyZB = -(Pi G ρ/(6 r3^3)) ((1/l)(r^4 + r^2 r3^2 + 4 r3^4 - r r3 (r^2 + 4 r3^2) Cos[θ3] -
          r^2 (3 r^2 + r3^2) Cos[2 θ3] + 3 r^3 r3 Cos[3 θ3]) + 6 r^3 Sin[θ3]^2 * Cos[θ3] * Log[B]);

In[•]:= VzzZB = (Pi G ρ/(3 r3^3)) ((1/l)(r^4 + r^2 r3^2 + 4 r3^4 - r r3 (r^2 + 4 r3^2) Cos[θ3] -
          r^2 (3 r^2 + r3^2) Cos[2 θ3] + 3 r^3 r3 Cos[3 θ3]) + 6 r^3 Sin[θ3]^2 * Cos[θ3] * Log[B]);

In[•]:= FullSimplify[VxxZB + VyyZB + VzzZB]
Out[•]=
0
```

5. Table 1

```
In[•]:= Clear["Global`*"];

In[•]:= VxxLS = G ρ Cos[2 λ - λ1 - λ2] Sin[λ2 - λ1] (Log[(r - r1)/(r - r2)] + Log[(r + r2)/(r + r1)]) -
          (2 G ρ/(3 r3^3)) (r3^3 (λ2 - λ1) + r3 Cos[2 λ - λ1 - λ2] Sin[λ2 - λ1] (3 r^2 + r3^2));

In[•]:= VyyLS = -G ρ Cos[2 λ - λ1 - λ2] Sin[λ2 - λ1] (Log[(r - r1)/(r - r2)] + Log[(r + r2)/(r + r1)]) -
          (2 G ρ/(3 r3^3)) (r3^3 (λ2 - λ1) - r3 Cos[2 λ - λ1 - λ2] Sin[λ2 - λ1] (3 r^2 + r3^2));

In[•]:= VzzLS = (4 G ρ (λ2 - λ1)/(3 r3^3)) r3^3;

In[•]:= VxxzLS = (2 G ρ r2^3/r^4) ((λ2 - λ1) - r2^2 Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1]/(r^2 - r2^2)) -
          (2 G ρ r1^3/r^4) ((λ2 - λ1) - r1^2 Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1]/(r^2 - r1^2));

In[•]:= VyyzLS = (2 G ρ r2^3/r^4) ((λ2 - λ1) + r2^2 Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1]/(r^2 - r2^2)) -
          (2 G ρ r1^3/r^4) ((λ2 - λ1) + r1^2 Cos[2 λ - λ1 - λ2] * Sin[λ2 - λ1]/(r^2 - r1^2));

In[•]:= VzzzLS = -(4 G ρ (λ2 - λ1)/r^4) (r2^3 - r1^3);

In[•]:= FullSimplify[VxxLS + VyyLS + VzzLS]
Out[•]=
0

In[•]:= FullSimplify[VxxzLS + VyyzLS + VzzzLS]
Out[•]=
0
```

6. Table 2

```
In[ ]:= Clear["Global`*"];
```

$$\text{In[ ]:= } \text{VxxSH} = -\frac{4\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{3\,\text{r}^3};$$

$$\text{In[ ]:= } \text{VyySH} = -\frac{4\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{3\,\text{r}^3};$$

$$\text{In[ ]:= } \text{VzzSH} = \frac{8\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{3\,\text{r}^3};$$

$$\text{In[ ]:= } \text{VxxzSH} = \frac{4\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{\text{r}^4};$$

$$\text{In[ ]:= } \text{VyyzSH} = \frac{4\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{\text{r}^4};$$

$$\text{In[ ]:= } \text{VzzzSH} = -\frac{8\,\text{Pi}\,\text{G}\,\rho\,\left(\text{r2}^3 - \text{r1}^3\right)}{\text{r}^4};$$

```
In[ ]:= FullSimplify[VxxSH + VyySH + VzzSH]
Out[ ]=
0
```

```
In[ ]:= FullSimplify[VxxzSH + VyyzSH + VzzzSH]
Out[ ]=
0
```